

Stock Price Prediction Minor Project Report

Chronological Evaluation with Ridge Regression

Prepared by Siddhant

Minor Project

Dataset AAPL daily closing prices

Verified period 2020-01-02 to 2024-12-31

Repository <https://github.com/Siddhantkr07/ML-Minor>

Abstract

This project evaluates whether a regression model can forecast the next trading day's closing price from recent price history. Daily AAPL data from 2020-01-02 to 2024-12-31 was retrieved through Yahoo Finance using yfinance. The model creates lag, return, and rolling-window features and is evaluated on the newest 20 percent of eligible observations. Ridge Regression achieved MAE 2.3132, RMSE 3.0384, and R-squared 0.9858. The naive persistence baseline achieved a lower MAE of 2.1306, so it remained the stronger benchmark for this run. The project is an educational forecasting experiment and not investment advice.

Problem Statement

The objective is to estimate the next trading day's closing price using only information available at the end of the current trading day. This is a time-series regression problem, so the model must be evaluated on a later period rather than on randomly shuffled rows.

Project Objectives

- Retrieve and validate historical daily closing-price data.
- Construct predictive features without using future prices as inputs.
- Train a reproducible Ridge Regression pipeline.
- Compare the model against a chronological holdout set and naive benchmark.
- Save the chart, metrics, predictions, dataset snapshot, and trained model for each run.

Dataset

The verified experiment uses Apple Inc. stock history with ticker AAPL. The data loader accepts Yahoo Finance downloads or a local CSV with Date and Close columns. It sorts dates, retains the final occurrence of any duplicate date, converts close values to numeric form, removes invalid values, and requires positive closing prices.

Attribute	Verified value
Data source	Yahoo Finance through yfinance
Ticker	AAPL
Observed date range	2020-01-02 to 2024-12-31
Trading-day observations	1258
Close price range	56.09 to 259.02
Mean close price	154.11

Feature Engineering

For a row dated t , the target is the closing price at t plus one trading day. Every input feature is calculated from the current or earlier closing prices. This construction prevents the model from reading the future target while training or evaluating.

Feature group	Variables used
Current price	Current closing price
Lagged price	1, 2, 3, 5, 10, and 20 trading-day lags
Momentum	Daily percentage return
Rolling statistics	5, 10, and 20-day means and standard deviations
Relative price	Distance from each rolling mean

Model and Evaluation Design

The model pipeline applies median imputation, standard scaling, and Ridge Regression with alpha equal to 1.0. The latest 20 percent of completed rows is reserved as a chronological test set. The baseline predicts that the next close equals the current close.

Experiment setting	Verified value
Forecast horizon	1 trading day
Training rows	989
Test rows	248
Primary model	Median imputation + StandardScaler + Ridge Regression
Comparison model	Naive persistence baseline

Results

Results below come from the final chronological holdout period. Lower MAE and RMSE are better; higher R-squared is better.

Model	MAE	RMSE	R-squared
Ridge Regression	2.3132	3.0384	0.9858
Naive persistence	2.1306	2.8970	0.9871

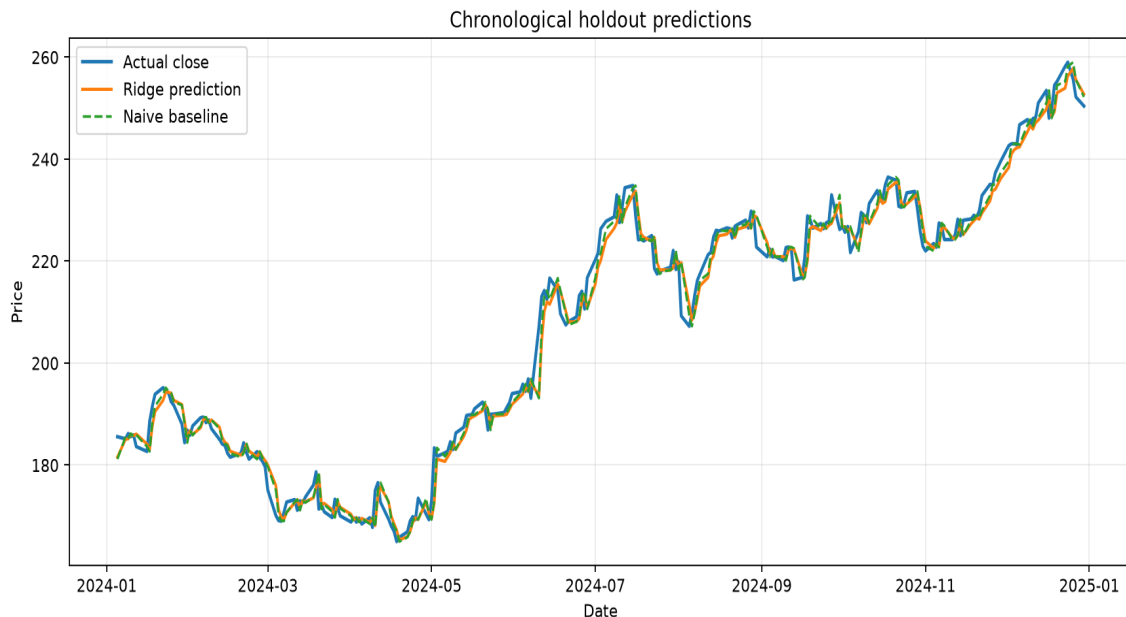


Figure 1. Chronological holdout predictions for AAPL.

Discussion

Ridge Regression follows the broad movement of the actual closing-price series and produces a high R-squared on this holdout period. However, the naive persistence baseline achieves lower MAE and RMSE and slightly higher R-squared. Therefore, the Ridge model did not improve on the simple current-close prediction for this particular experiment. This comparison is important because it prevents overstatement of the forecast's practical performance.

Limitations

- Only historical closing prices are used; news, earnings, fundamentals, macroeconomic data, and intraday signals are excluded.
- The experiment covers one asset and one holdout period, so results may not generalize to other stocks or market regimes.
- Trading costs, portfolio risk, and investment suitability are not assessed.
- The output is not financial advice and must not be used as an automated trading signal.

Reproducibility

The repository includes source code, automated tests, the dataset documentation, and the model card. Run the following commands from the project root after installing dependencies:

```
$env:PYTHONPATH = "src"
.\.venv\Scripts\python.exe -m stock_predictor.cli train --ticker AAPL --start 2020-01-01
--end 2025-01-01
```

Conclusion

This project provides a reproducible next-day stock-price forecasting workflow with data validation, leakage-aware features, chronological evaluation, automated tests, and documented outputs. The verified AAPL experiment shows that Ridge Regression can track price movements, while the naive baseline remains stronger on the selected holdout metrics. Future work should evaluate walk-forward validation, multiple assets, and carefully validated additional predictors.

References

- Yahoo Finance data accessed through yfinance. <https://github.com/ranaroussi/yfinance>
- scikit-learn Ridge Regression documentation.
https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Ridge.html
- Hyndman, R. J., and Athanasopoulos, G. Forecasting Principles and Practice.
<https://otexts.com/fpp3/tscv.html>
- Project repository. <https://github.com/Siddhantkr07/ML-Minor>